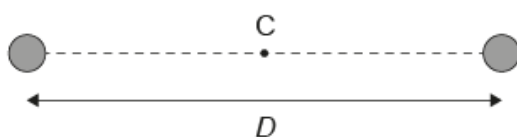


Topic D HL - Paper 2 - Study Guide [178 marks]

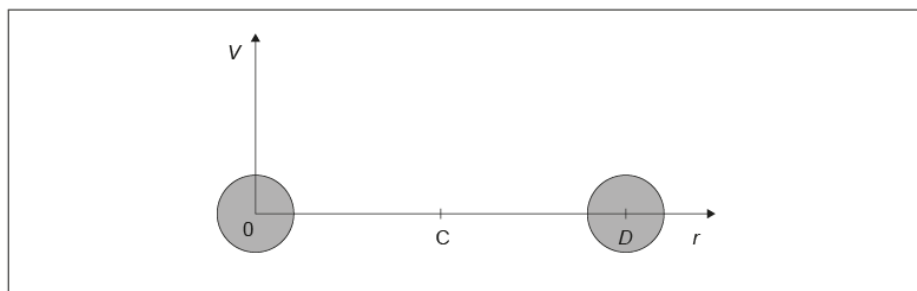
1. [Maximum mark: 11]

23M.2.HL.TZ1.6

- (a) The centres of two identical fixed conducting spheres each of charge $+Q$ are separated by a distance D . C is the midpoint of the line joining the centres of the spheres.



- (a.i) Sketch, on the axes, how the electric potential V due to the two charges varies with the distance r from the centre of the left charge. No numbers are required. Your graph should extend from $r = 0$ to $r = D$.



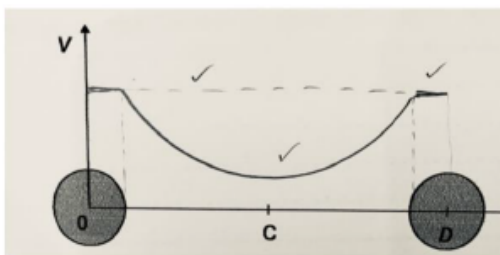
[3]

Markscheme

Constant, non-zero within spheres ✓

A clear, non-zero positive minimum at C ✓

Symmetric bowl shaped up curved shape in between ✓



Do **not** allow a bowl shaped down curve for **MP3**.

Examiners report

This graph proved challenging for students. Very few recognized that there would be a non-zero, constant potential inside the spheres. Some recognized that there would be a minimum at C, although many had the graph touch the axis at C which resulted in a loss of the second marking point.

- (a.ii) Calculate the work done to bring a small charge q from infinity to point C.

Data given:

$$Q = 2.0 \times 10^{-3} \text{ C},$$

$$q = 4.0 \times 10^{-9} \text{ C}$$

$$D = 1.2 \text{ m}$$

[2]

Markscheme

$$V = 2 \times \frac{8.99 \times 10^9 \times 2.0 \times 10^{-3}}{0.60} = 6.0 \times 10^7 \text{ «V»} \checkmark$$

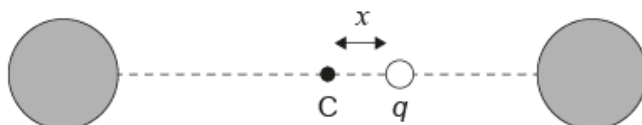
$$W = «qV = 6.0 \times 10^7 \times 4.0 \times 10^{-9} =» 0.24 \text{ «J»} \checkmark$$

Allow **ECF** from **MP1**

Examiners report

Few students were able to successfully calculate the work done in moving a charge to point C. Many calculated the work for only one of the charges, or did not divide the distance between the charges in half. ECF was awarded to many students for these common mistakes.

- (b) The small positive charge q is placed a distance x to the right of C. The distance x is very small compared to D .



- (b.i) The magnitude of the net force on q is given by $\frac{32kQq}{D^3} x$. Explain why the charge q will execute simple harmonic oscillations about C.

[2]

Markscheme
The restoring force/acceleration is opposite to the displacement/towards equilibrium / OWTTE ✓ and proportional to displacement from equilibrium / OWTTE ✓ <i>Allow discussions based on the diagram (such as towards C for towards equilibrium).</i> Accept $F \propto x$ OR $a \propto x$ for MP2

Examiners report
Examiners were looking for students to apply the key concepts of simple harmonic motion to this situation. Many students were able to earn one of the marks (the restoring force towards C or the fact that the force/acceleration is proportional to displacement), but many did not discuss both points for full marks.

(b.ii) The mass of the charge q is 0.025 kg.

Calculate the angular frequency of the oscillations using the data in (a)(ii) and the expression in (b)(i).

[2]

Markscheme
$\omega = \sqrt{\frac{32kQq}{mD^3}} \text{ OR use of } F = m\omega^2 r \text{ OR } F = 1.33x \text{ OR } a = 53.3x \checkmark$ $\ll \sqrt{\frac{32 \times 8.99 \times 10^9 \times 2.0 \times 10^{-3} \times 4.0 \times 10^{-9}}{0.025 \times 1.2^3}} \gg = 7.299 \text{ «s}^{-1}\text{»}$

Examiners report
This question looked for the angular frequency of the charge. A surprising number of students left this question blank. Some were able to earn the first marking point for setting up an intermediate calculation, but few were able to successfully calculate the final result.

(c) The charges Q are replaced by neutral masses M and the charge q by a neutral mass m . The mass m is displaced away from C by a small distance x and released. Discuss whether the motion of m will be the same as that of q .

[2]

Markscheme

the net force will no longer be a restoring force/directed towards equilibrium

OR

the gravitational force is attractive/neutral mass would be pulled towards larger masses/**OWTTE**

✓

«and so» no, motion will not be the same/no longer be SHM / **OWTTE** ✓

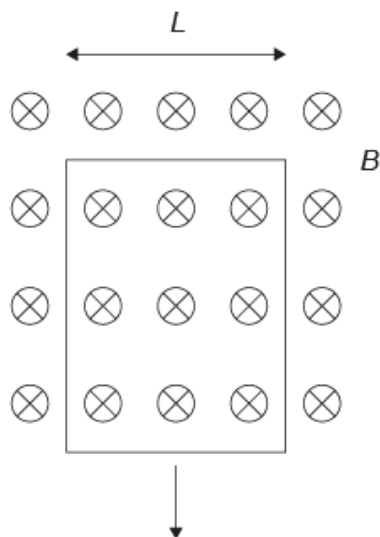
Examiners report

Students did not fully appreciate the differences between the original situation with two charges and the situation with two masses. Most recognized that the motion would not be the same for the first marking point, but many failed to consider the impact of the attractive force of gravity. Many simply focused on the "neutral" part, and thus said there would be no force, or that the force would be too weak to cause SHM.

2. [Maximum mark: 12]

23M.2.HL.TZ1.7

A vertical rectangular loop of conducting wire is dropped in a region of horizontal magnetic field. The diagram shows the loop as it leaves the region of the magnetic field.



- (a) Explain, by reference to Faraday's law of electromagnetic induction, why there is an electromotive force (emf) induced in the loop as it leaves the region of magnetic field.

[2]

Markscheme

The induced emf is equal/proportional/related to the «rate of» change of «magnetic» flux/flux linkage ✓

Flux is changing because the area pierced/enclosed by magnetic field lines changes «decreases»

OR

Flux is changing because the loop is leaving/moving out of the «magnetic» field. ✓

*Need to see a connection between the EMF and change in flux for **MP1**.*

*Need to see a connection between the area changing or leaving the field and the change in flux for **MP2***

Examiners report

Examiners were looking for students to link two concepts: first, a changing magnetic flux leads to an EMF, and that the flux is changing because either area in the magnetic field was decreasing or because the loop was leaving the field. One common issue students had was keeping the difference between magnetic field and magnetic flux straight in their answers. A common incorrect attempt was to state that the EMF is induced because the loop was moving in a magnetic field. It should be noted that the original examination paper incorrectly expanded EMF as electromagnetic force rather than electromotive force.

- (b) Just before the loop is about to completely exit the region of magnetic field, the loop moves with constant terminal speed v .

The following data is available:

Mass of loop	$m = 4.0 \text{ g}$
Resistance of loop	$R = 25 \text{ m}\Omega$
Width of loop	$L = 15 \text{ cm}$
Magnetic flux density	$B = 0.80 \text{ T}$

Determine, in m s^{-1} the terminal speed v .

[4]

Markscheme

$$mg = BIL \text{ OR } I = 0.33 \text{ «A» } \checkmark$$

$$BvL = IR \text{ OR } \mathcal{E} = 8.25 \times 10^{-3} \text{ «V» OR } \mathcal{E} = 0.12v \checkmark$$

Combining results to get $v = \frac{mgR}{B^2L^2}$ ✓

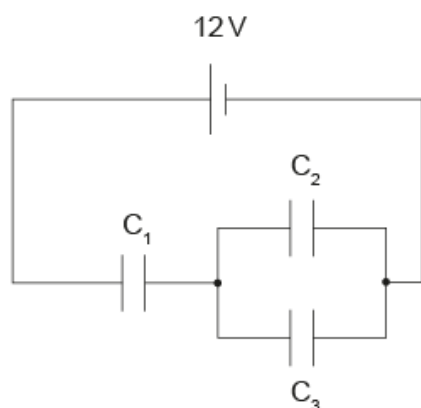
$$v = \ll \frac{0.040 \times 9.81 \times 0.025}{0.80^2 \times 0.15^2} \gg = 0.068 \ll \text{m s}^{-1} \gg$$

Allow **ECF** between steps if clear work is shown.

Examiners report

This was a poorly done question. There were many attempts to write random equations down in hopes of earning a mark, but few successful attempts. Very few students appreciated the fact that the magnetic force acting on the loop had to be equal to the weight, which greatly limited their ability to earn marks.

- (c) Three capacitors $C_1 = 3.0 \mu\text{F}$, $C_2 = 2.0 \mu\text{F}$ and $C_3 = 4.0 \mu\text{F}$ are connected to a cell of emf 12 V and negligible internal resistance. The capacitors are initially uncharged.



Calculate

- (c.i) the total capacitance of the circuit.

[2]

Markscheme

The 2 in parallel give a total of $6.0 \ll \mu\text{F} \gg$ ✓

The total is $\ll \left(\frac{1}{3} + \frac{1}{6} \right)^{-1} \gg = 2.0 \ll \mu\text{F} \gg$ ✓

Allow **ECF** from **MP1**

Accept other powers of 10 for capacitances with proper unit included.

Examiners report

Students were generally successful on this item, with most correctly calculating the equivalent capacitance of the parallel section and then using this to do a proper series calculation. Common mistakes included reversing these calculations (essentially treating the capacitors as resistors), basic mathematical errors, and forgetting to take the reciprocal at the end of the series calculation.

(c.ii) the total energy stored in the three capacitors.

[1]

Markscheme

$$E = \frac{1}{2} CV^2 = \frac{1}{2} \times 2.0 \times 10^{-6} \times 12^2 = 1.44 \times 10^{-4} \text{ «J»} \checkmark$$

Allow ECF from **c(i)** (=72 × **c(ii)**)

Examiners report

This was generally well done. The biggest mistake students made was in not converting the capacitance to farads leading to power of ten errors. While a unit mark was not awarded here, the assumption is that a unitless answer is in proper MKS units, so an answer of "144" would not be awarded a mark for a power of ten error (however, 144 with a proper unit of μJ would earn the mark).

(c.iii) the charge on C_2 .

[3]

Markscheme

ALTERNATE 1

Voltage across C_2 is half that across C_1 ✓

So voltage across C_2 is 4.0 V ✓

Charge is $\llcorner C_2 V_2 = 2.0 \times 10^{-6} \times 4.0 \gg 8.0 \times 10^{-6} \text{ «C»} \checkmark$

ALTERNATE 2

Charge on C_1 is $\llcorner C_1 V_T = 2.0 \times 10^{-6} \times 12 \gg 24 \text{ «}\mu\text{C}\gg \checkmark$

So voltage across C_1 is $\llcorner \frac{24}{3.0} \gg 8.0 \text{ «V»} \checkmark$

Charge on C_2 is « $C_2V_2 = 2.0 \times 10^{-6} \times 4.0$ » 8.0×10^{-6} «C» ✓

ALTERNATE 3

« $C_3 = 2C_2$ leading to » $q_3 = 2q_2$ ✓

Total charge in parallel = « $q_2 + q_3 = q_2 + 2q_2$ » $3q_2$ ✓

$3q_2 = 24$ leading to $q_2 = 8 \times 10^{-6}$ «C» ✓

ECF for MP3 allowed in ALT 1 and ALT 2

Examiners report

Students struggled with this question. While some were able to successfully calculate the charge on C_2 , most made many errors along the way. Some students calculated the total charge in the circuit, but assumed that this was also the charge on C_2 . This was awarded one mark.

3. [Maximum mark: 11]

23M.2.HL.TZ2.6

A moon M orbits a planet P . The gravitational field strength at the surface of P due to P is g_P .

The gravitational field strength at the surface of M due to M is g_M .

For M and P : $\frac{\text{radius of } M}{\text{radius of } P} = 0.27$ and $\frac{\text{mass of } M}{\text{mass of } P} = 0.055$

(a) Determine $\frac{g_M}{g_P}$.

[2]

Markscheme

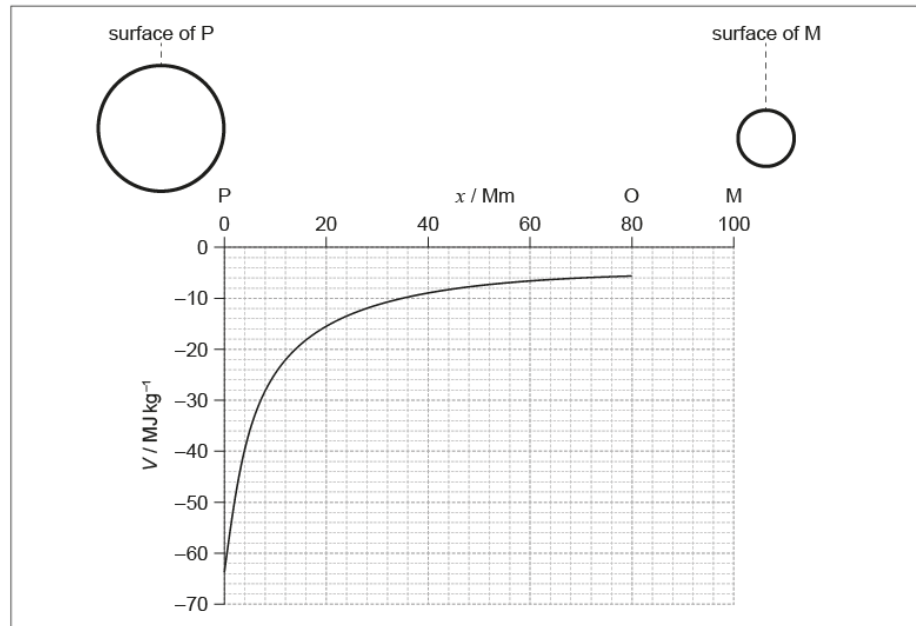
Work using $g \propto \frac{m}{r^2}$ ✓

$$\frac{g_M}{g_P} = \frac{m_M}{m_P} \left(\frac{r_P}{r_M} \right)^2 = 0.75 \quad \checkmark$$

Examiners report

This part was generally well answered.

- (b) Point O lies on the line joining the centre of M to the centre of P.



The graph shows the variation of gravitational potential V with distance x from the surface of P to O.

The gradient of the graph is zero at point O.

- (b.i) State and explain the magnitude of the resultant gravitational field strength at O.

[2]

Markscheme

$g = 0$ ✓

As $g \ll -\frac{\Delta V_g}{\Delta r}$ which is the gradient of the graph

OR

As the force of attraction/field strength of P and M are equal ✓

Examiners report

Many students related the gradient of the graph to the gravitational field strength and so concluded it was zero. It was also acceptable to talk about the equal forces of attraction between P and M.

- (b.ii) Outline why the graph between P and O is negative.

[2]

Markscheme

The gravitational field is attractive so that energy is required «to move away from P» ✓

the gravitational potential is defined as 0 at ∞ , (the potential must be negative) ✓

Examiners report

Many students quoted a definition of gravitational potential but didn't go further.

- (b.iii) Show that the gravitational potential V_P at the surface of P due to the mass of P is given by $V_P = -g_P R_P$ where R_P is the radius of the planet.

[2]

Markscheme

$$V_P = -\frac{GM}{R_P} \text{ AND } g_P = \frac{GM}{R_P^2} \text{ (at surface) } \checkmark$$

Suitable working and cancellation of G and M seen ✓

$$V_P = -g_P R_P$$

Must see negative sign

Examiners report

It was common to see answers that started with $g = -\Delta V / \Delta r$ and then cancelled the deltas.

- (b.iv) The gravitational potential due to the mass of M at the surface of P can be assumed to be negligible.

Estimate, using the graph, the gravitational potential at the surface of M due to the mass of M.

[2]

Markscheme

$$\ll \frac{V_M}{V_P} = \frac{g_M R_M}{g_P R_P} = 0.75 \times 0.27 \gg = 0.20 \checkmark$$

$$V_M = \ll -6.4 \times 10^7 \times 0.2 \gg \ll - \gg 1.3 \times 10^7 \text{ J kg}^{-1} \gg \checkmark$$

Examiners report

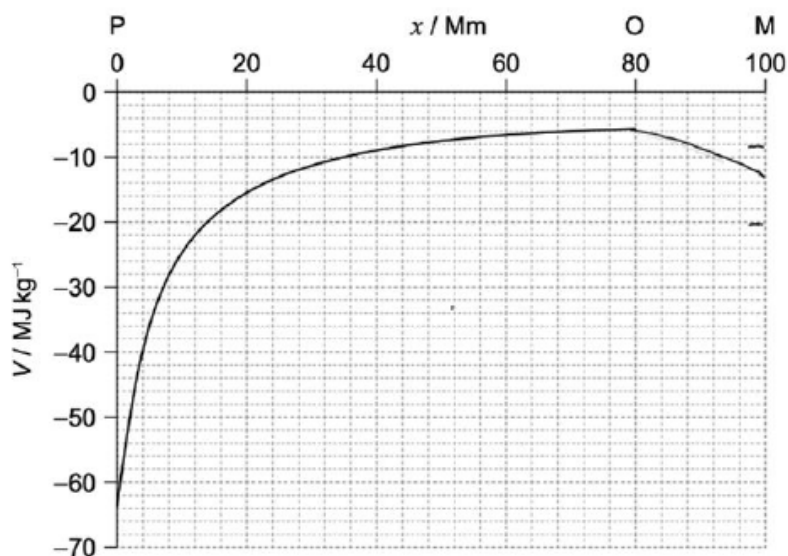
Few students realised the link between this part of the question and the previous one, but those that did tended to get the correct answer. Common incorrect answers were -6 MJ kg^{-1} and -64 MJ kg^{-1} .

(b.v) Draw on the axes the variation of gravitational potential between O and M.

[1]

Markscheme

Line always negative, of suitable shape and end point below -8 and above -20 unless awarding *ECF* from *b(iv)* ✓



Examiners report

Many students seemed to miss this part of the question as the graph was often left untouched. A range of end points was accepted for the line on the graph, but error carried forward was also allowed from *b(iv)* however unlikely the end result.

4. [Maximum mark: 9]

22M.2.HL.TZ1.7

Two identical positive point charges X and Y are placed 0.30 m apart on a horizontal line. O is the point midway between X and Y. The charge on X and the charge on Y is $+4.0 \mu\text{C}$.

(a) Calculate the electric potential at O.

[3]

Markscheme

use of $\frac{kQ}{r}$ ✓

$$\frac{(8.99 \times 10^9)(4 \times 10^{-6})}{0.15} \text{ OR } 240 \text{ «kV» for one charge calculated ✓}$$

480 «kV» for both ✓

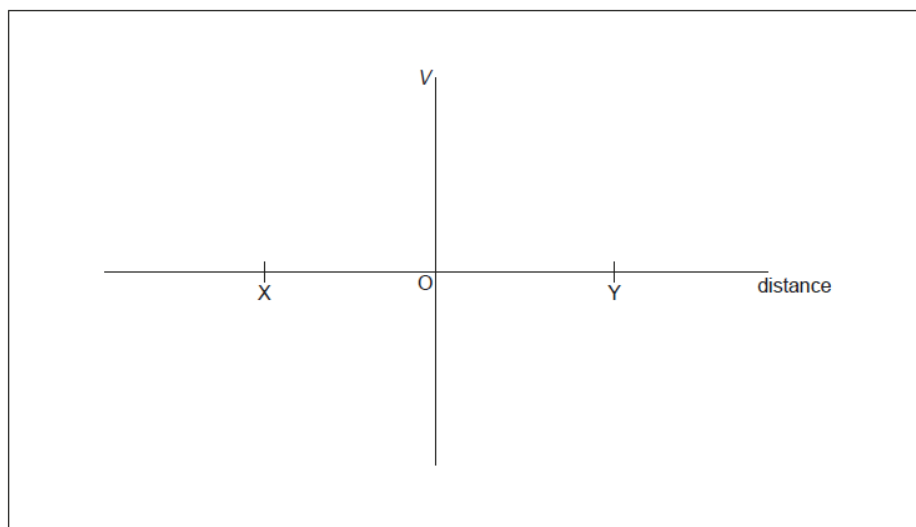
MP1 can be seen or implied from calculation.

Allow **ECF** from **MP2** for **MP3**.

Examiners report

This question was generally well approached. Two common errors were either starting with the wrong equation (electric potential energy or Coulomb's law) or subtracting the potentials rather than adding them.

- (b) Sketch, on the axes, the variation of the electric potential V with distance between X and Y.

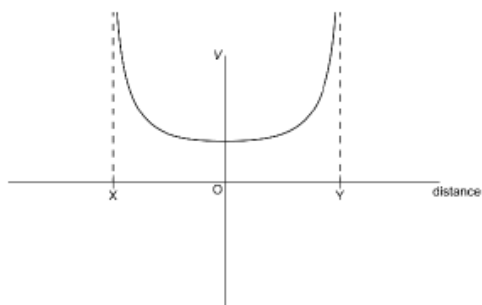


[2]

Markscheme

symmetric curve around 0 with potential always positive, “bowl shape up” and curve not touching the horizontal axis. ✓

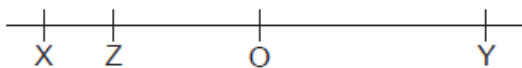
clear asymptotes at X and Y ✓



Examiners report

Very few candidates drew a graph that was awarded two marks. Many had a generally correct shape, but common errors were drawing the graph touching the x-axis at O and drawing a general parabola with no clear asymptotes.

A positive charge Z is released from rest 0.010 m from O on the line between X and Y. Z then begins to oscillate about point O.



(c.i) Identify the direction of the resultant force acting on Z as it oscillates.

[2]

Markscheme

force is towards O ✓

always ✓

Examiners report

Many candidates were able to identify the direction of the force on the particle at position Z, but a common error was to miss that the question was about the direction as the particle was oscillating. Examiners were looking for a clear understanding that the force was always directed toward the equilibrium position, and not just at the moment shown in the diagram.

(c.ii) Deduce whether the motion of Z is simple harmonic.

[2]

Markscheme

ALTERNATIVE 1

motion is not SHM ✓

«because SHM requires force proportional to r and» this force depends on $\frac{1}{r^2}$ ✓

ALTERNATIVE 2

motion is not SHM ✓

energy-distance «graph must be parabolic for SHM and this» graph is not parabolic ✓

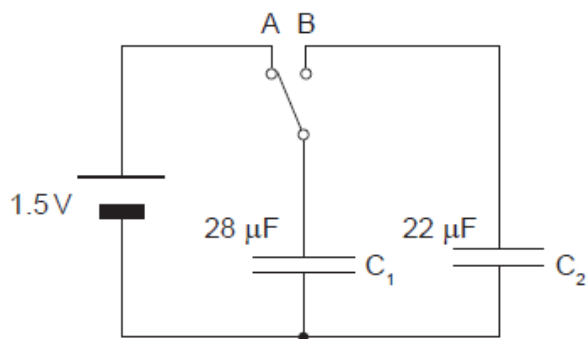
Examiners report

This was a challenging question for candidates. Most simply assumed that because the charge was oscillating that this meant the motion was simple harmonic. Some did recognize that it was not, and most of those candidates correctly identified that the relationship between force and displacement was an inverse square.

5. [Maximum mark: 11]

22M.2.HL.TZ1.8

Two capacitors C_1 and C_2 of capacitance $28 \mu\text{F}$ and $22 \mu\text{F}$ respectively are connected in a circuit with a two-way switch and a cell of emf 1.5 V with a negligible internal resistance. The capacitors are initially uncharged. The switch is then connected to position A.



(a) Show that the charge stored on C_1 is about 0.04 mC .

[1]

Markscheme

$$Q = CV = 1.5 \times 28 \times 10^{-6} = 0.042 \text{ «mC» } \checkmark$$

Award **MP** for full replacement or correct answer to at least 2 significant figures.

Examiners report

This was a "show that" question, and it was very well done by most candidates.

The switch is moved to position B.

(b.i) Calculate the energy transferred from capacitor C_1 .

[4]

Markscheme

$$E_i = \frac{1}{2} \times (28 \times 10^{-6})(1.5)^2 = 3.15 \times 10^{-5} \text{ «J» } \checkmark$$

total capacitance = 50 « μ F»

OR

$$\text{pd} = \left\langle \frac{42 \times 10^{-6}}{50 \times 10^{-6}} \right\rangle = 0.84 \text{ «V»}$$

OR

charge on C_1 after switch moved to B = 0.0235 «mC» $\checkmark$

$$E_f = \frac{1}{2} \times (28 \times 10^{-6})(0.84)^2 = 9.9 \times 10^{-6} \text{ «J» } \checkmark$$

$$\text{energy lost} = 3.2 \times 10^{-5} - 9.9 \times 10^{-6} = 22 \text{ «}\mu\text{J» } \checkmark$$

Examiners report

This question was challenging for many candidates. A large number successfully calculated the initial energy of C_1 , but then seemed confused about the next steps. Few candidates successfully calculated the energy in C_1 after the switch was closed. There was an ECF

opportunity for candidates who recognized that the final answer was the difference between these two values.

(b.ii) Explain why the energy gained by capacitor C_2 differs from your answer in (b)(i).

[1]

Markscheme

energy transferred to electromagnetic radiation «to environment»

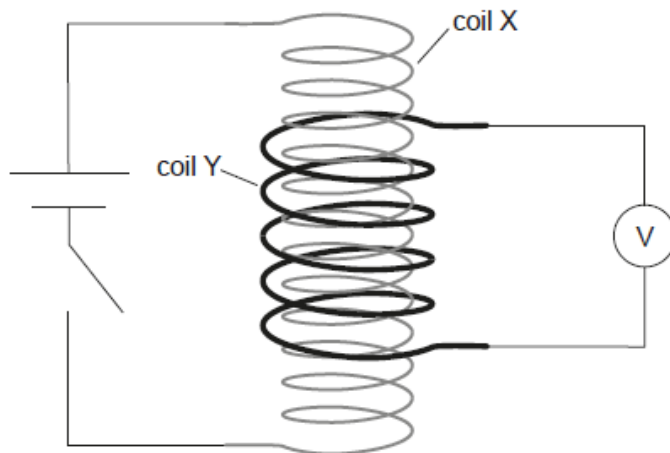
OR

energy is transferred as thermal energy / heat «to circuit components» ✓

Examiners report

This question used an "explain" command term, so examiners were looking for more than a generic "energy was lost". Candidates needed to specify a form of energy that was lost (such as thermal energy) for the mark. A very common incorrect response was simply stating that the difference was due to the capacitors having a different capacitance.

A cell is now connected by a switch to a coil X. A second coil Y of cross-sectional area 6.4 cm^2 with 5 turns is looped around coil X and connected to an ideal voltmeter.



(c.i) The switch is closed at time $t = 0$. Explain how the voltmeter reading varies after the switch is closed.

[2]

Markscheme

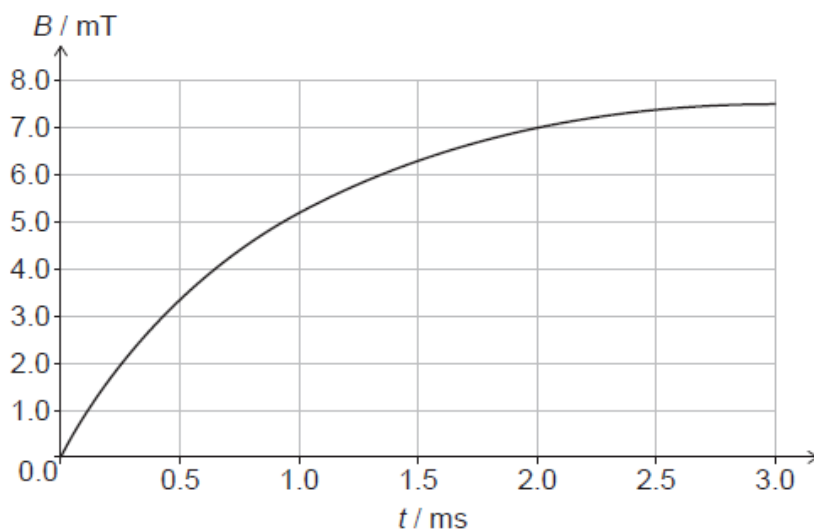
initial deflection by voltmeter falling to zero reading ✓

emf is induced «only» while the field / flux is changing ✓

Examiners report

This question was well answered by some candidates who recognized that it was an electromagnetic induction question and understood that eventually the current in coil X would hit a steady state and the voltmeter reading would return to zero. Common issues were candidates thinking that the potential would fluctuate in a manner similar to an alternating current, candidates discussing this more as a transformer, and candidates who missed that there were two separate coils and wrote responses suggesting that a simple circuit had been formed and the voltmeter would read the potential of the cell.

The graph shows the variation with t of the magnetic flux density B in coil Y.



(c.ii) Determine the average emf induced across coil Y in the first 3.0 ms.

[3]

Markscheme

attempted use of $\varepsilon = \frac{\Delta\Phi}{\Delta t}$ *OR* $NA \frac{\Delta B}{\Delta t}$ ✓

$$\varepsilon = \frac{5 \times 7.5 \times 10^{-3} \times 6.4 \times 10^{-4}}{3 \times 10^{-3}} \quad \checkmark$$

8.0 «mV» ✓

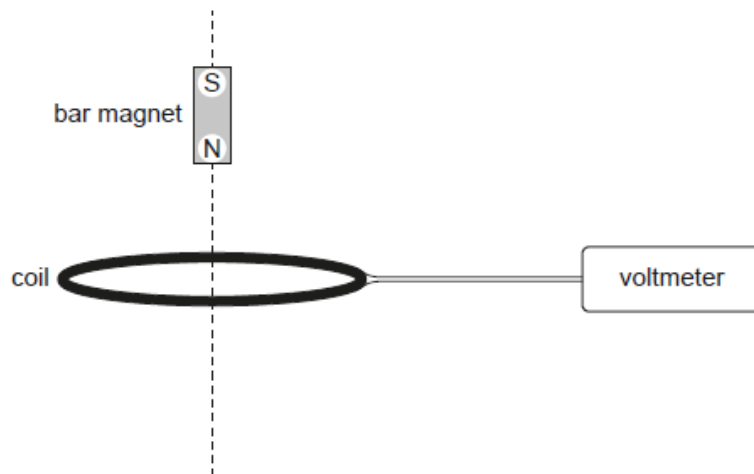
Examiners report

This question was well approached with most candidates recognizing that this was a Faraday's law question. Many made an attempt to use the correct equation, but common errors were choosing incorrect values from the graph and incorrectly converting the given area. Examiners were generous with ECF for candidates who clearly showed work leading to an incorrect result.

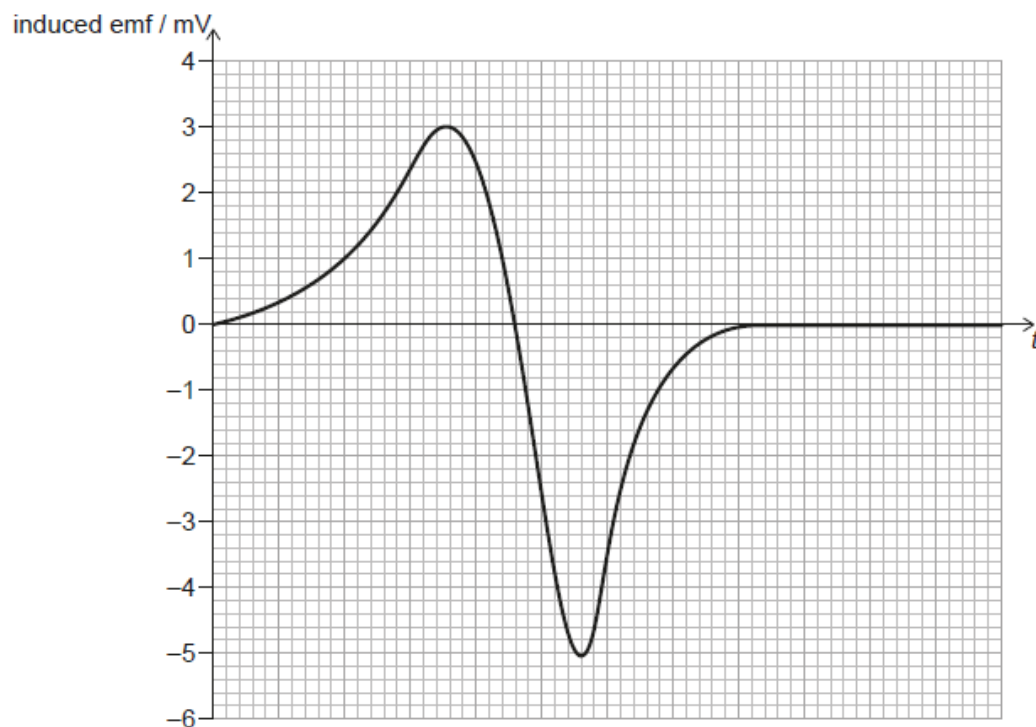
6. [Maximum mark: 10]

22M.2.HL.TZ2.8

A fixed horizontal coil is connected to an ideal voltmeter. A bar magnet is released from rest so that it falls vertically through the coil along the central axis of the coil.



The variation with time t of the emf induced in the coil is shown.



- (a.i) Write down the maximum magnitude of the rate of change of flux linked with the coil.

[1]

Markscheme

«→» 5.0 «mV» **OR** 5.0×10^{-3} «V» ✓

Accept 5.1

Examiners report

ai

Well answered, with wrong answers stating 8 for the difference or 3 without realising that the sign does not matter.

aii

Very few candidates managed to get the correct fundamental SI unit for V. All kinds of errors were observed, from power errors to the use of C as a fundamental unit instead of A.

bi) Most scored best by marking using an alternative method introduced to the markscheme in standardisation. There were some confused and vague comments. Clear, concise answers were

rare.

bii) It was common to see conservation of energy invoked here with suggestions that energy was the area under the graph. Many candidates described the shapes to explain why the areas were the same rather than talking about the physics e.g. one peak is short and fat and the other is tall and thin so they balance out.

c) A surprising number didn't pick up on the fact that the magnet would be moving slower. As a result, they discussed everything happening sooner, i.e. the interaction with the magnet and the coil, and that led onto things happening quicker so peaks being bigger.

(a.ii) State the fundamental SI unit for your answer to (a)(i).

[1]

Markscheme

$\text{kg m}^2 \text{A}^{-1} \text{s}^{-3}$ ✓

(b.i) Explain why the graph becomes negative.

[3]

Markscheme

ALTERNATIVE 1

Flux linkage is represented by magnetic field lines through the coil ✓

when magnet has passed through the coil / is moving away ✓

flux «linkage» is decreasing ✓

suitable comment that it is the opposite when above ✓

when the magnet goes through the midpoint the induced emf is zero ✓

ALTERNATIVE 2

reference to / states Lenz's law ✓

when magnet has passed through the coil / is moving away ✓

«coil attracts outgoing S pole so» induced field is downwards ✓

before «coil repels incoming N pole so» induced field is upwards

OR

induced field has reversed ✓

when the magnet goes through the midpoint the induced emf is zero ✓

OWTTE

- (b.ii) Part of the graph is above the t -axis and part is below. Outline why the areas between the t -axis and the curve for these two parts are likely to be the same.

[2]

Markscheme

area represents the total change in flux «linkage» ✓

the change in flux is the same going in and out ✓

«when magnet is approaching» flux increases to a maximum ✓

«when magnet is receding» flux decreases to zero ✓

«so areas must be the same»

- (c) Predict the changes to the graph when the magnet is dropped from a lower height above the coil.

[3]

Markscheme

magnet moves slower ✓

overall time «for interaction» will be longer ✓

peaks will be smaller ✓

areas will be the same as before ✓

*Allow a graphical interpretation for **MP2** as “graph more spread out”*

7. [Maximum mark: 9]

22M.2.HL.TZ2.7

A metal sphere is charged positively and placed far away from other charged objects. The electric potential at a point on the surface of the sphere is 53.9 kV.

(a) Outline what is meant by electric potential at a point.

[2]

Markscheme

the work done per unit charge ✓

In bringing a small/point/positive/test «charge» from infinity to the point ✓

*Allow use of energy per unit charge for **MP1***

Examiners report

a) Well answered.

b) Generally, well answered, but there were quite a few using $r + 2.8$.

ci) Very few had problems to recognize the perpendicular angle

cii) Good simple calculation

ciii) Many had a good go at this, but a significant number tried to answer it based on forces.

(b) The electric potential at a point a distance 2.8 m from the centre of the sphere is 7.71 kV. Determine the radius of the sphere.

[2]

Markscheme

use of $V/r = \text{constant}$ ✓

0.40 m ✓

*Allow **[1]** max if $r + 2.8$ used to get 0.47 m.*

*Allow **[2]** marks if they calculate Q at one potential and use it to get the distance at the other potential.*

A small positively charged object moves towards the centre of the metal sphere. When the object is 2.8 m from the centre of the sphere, its speed is 3.1 m s^{-1} . The mass of the object is 0.14 g and its charge is $2.4 \times 10^{-8} \text{ C}$.

- (c.i) Comment on the angle at which the object meets equipotential surfaces around the sphere. [1]

Markscheme
90° / perpendicular ✓

- (c.ii) Show that the kinetic energy of the object is about 0.7 mJ. [1]

Markscheme
$\frac{1}{2} \times 0.14 \times 10^{-3} \times 3.1^2$ OR 0.67 «mJ» seen ✓

- (c.iii) Determine whether the object will reach the surface of the sphere. [3]

Markscheme
«p.d. between point and sphere surface = » (53.9 kV – 7.71) «kV» OR 46.2 «kV» seen ✓ «energy required =» VQ « = 46 200 × 2.4 × 10⁻⁸ » = 1.11 mJ ✓ this is greater than kinetic energy so will not reach sphere ✓ MP3 is for a conclusion consistent with the calculations shown. Allow ECF from MP1

8. [Maximum mark: 8]

21M.2.HL.TZ1.2

A planet is in a circular orbit around a star. The speed of the planet is constant. The following data are given:

$$\begin{aligned} \text{Mass of planet} &= 8.0 \times 10^{24} \text{ kg} \\ \text{Mass of star} &= 3.2 \times 10^{30} \text{ kg} \\ \text{Distance from the star to the planet } R &= 4.4 \times 10^{10} \text{ m.} \end{aligned}$$

- (a) Explain why a centripetal force is needed for the planet to be in a circular orbit. [2]

Markscheme

«circular motion» involves a changing velocity ✓

«Tangential velocity» is «always» perpendicular to centripetal force/acceleration ✓

there must be a force/acceleration towards centre/star ✓

without a centripetal force the planet will move in a straight line ✓

(b) Calculate the value of the centripetal force.

[1]

Markscheme

$$F = \frac{(6.67 \times 10^{-11})(8 \times 10^{24})(3.2 \times 10^{30})}{(4.4 \times 10^{10})^2} = 8.8 \times 10^{23} \text{ «N» } \checkmark$$

A spacecraft is to be launched from the surface of the planet to escape from the star system. The radius of the planet is 9.1×10^3 km.

(c.i) Show that the gravitational potential due to the planet and the star at the surface of the planet is about $-5 \times 10^9 \text{ J kg}^{-1}$.

[3]

Markscheme

$$V_{\text{planet}} = \text{«-»} \frac{(6.67 \times 10^{-11})(8 \times 10^{24})}{9.1 \times 10^6} = \text{«-»} 5.9 \times 10^7 \text{ «J kg}^{-1}\text{» } \checkmark$$

$$V_{\text{star}} = \text{«-»} \frac{(6.67 \times 10^{-11})(3.2 \times 10^{30})}{4.4 \times 10^{10}} = \text{«-»} 4.9 \times 10^9 \text{ «J kg}^{-1}\text{» } \checkmark$$

$$V_{\text{planet}} + V_{\text{star}} = \text{«-»} 4.9 \text{ «09»} \times 10^9 \text{ «J kg}^{-1}\text{» } \checkmark$$

Must see substitutions and not just equations.

(c.ii) Estimate the escape speed of the spacecraft from the planet–star system.

[2]

Markscheme

$$\text{use of } v_{\text{esc}} = \sqrt{2V} \checkmark$$

$$v = 9.91 \times 10^4 \text{ «m s}^{-1}\text{» } \checkmark$$

9. [Maximum mark: 7]

21M.2.HL.TZ2.10

The table gives data for Jupiter and three of its moons, including the radius r of each object.

Object	Mass/kg	r / m	Orbital radius around Jupiter/m
Jupiter	1.9×10^{27}	7.1×10^7	
Io	8.9×10^{22}	1.8×10^6	4.9×10^8
Ganymede	1.5×10^{23}	2.6×10^6	1.06×10^9
Callisto	1.1×10^{23}	2.4×10^6	1.88×10^9

- (a) Calculate, for the surface of Io, the gravitational field strength g_{Io} due to the mass of Io. State an appropriate unit for your answer.

[2]

Markscheme

$$\ll \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 8.9 \times 10^{22}}{(1.8 \times 10^6)^2} = \gg 1.8 \checkmark$$

N kg^{-1} **OR** m s^{-2} $\checkmark$

A spacecraft is to be sent from Io to infinity.

- (b.i) Show that the $\frac{\text{gravitational potential due to Jupiter at the orbit of Io}}{\text{gravitational potential due to Io at the surface of Io}}$ is about 80.

[2]

Markscheme

$$\frac{1.9 \times 10^{27}}{4.9 \times 10^8} \text{ AND } \frac{8.9 \times 10^{22}}{1.8 \times 10^6} \text{ seen } \checkmark$$

$$\ll \frac{1.9 \times 10^{27} \times 1.8 \times 10^6}{4.9 \times 10^8 \times 8.9 \times 10^{22}} = \gg 78 \checkmark$$

For **MP1**, potentials can be seen individually or as a ratio.

- (b.ii) Outline, using (b)(i), why it is not correct to use the equation $\sqrt{\frac{2G \times \text{mass of Io}}{\text{radius of Io}}}$ to calculate the speed required for the spacecraft to reach infinity from the surface of Io.

[1]

Markscheme
«this is the escape speed for Io alone but» gravitational potential / field of Jupiter must be taken into account ✓
OWTTE

- (c) An engineer needs to move a space probe of mass 3600 kg from Ganymede to Callisto. Calculate the energy required to move the probe from the orbital radius of Ganymede to the orbital radius of Callisto. Ignore the mass of the moons in your calculation.

[2]

Markscheme
$-GM_{\text{Jupiter}} \left(\frac{1}{1.88 \times 10^9} - \frac{1}{1.06 \times 10^9} \right) = \ll 5.21 \times 10^7 \text{ J kg}^{-1} \gg \checkmark$ « multiplies by 3600 kg to get » $1.9 \times 10^{11} \ll \text{J} \gg \checkmark$ <i>Award [2] marks if factor of ½ used, taking into account orbital kinetic energies, leading to a final answer of $9.4 \times 10^{10} \ll \text{J} \gg$.</i> <i>Allow ECF from MP1</i> <i>Award [2] marks for a bald correct answer.</i>

10. [Maximum mark: 11]

21M.2.HL.TZ2.7

The primary coil of a transformer is connected to a 110 V alternating current (ac) supply. The secondary coil of the transformer is connected to a 15 V garden lighting system that consists of 8 lamps connected in parallel. Each lamp is rated at 35 W when working at its normal brightness. Root mean square (rms) values are used throughout this question.

- (a.i) The primary coil has 3300 turns. Calculate the number of turns on the secondary coil.

[1]

Markscheme

$$\ll \frac{15}{110} \times 3300 = \gg 450 \text{ «turns» } \checkmark$$

(a.ii) Determine the total resistance of the lamps when they are working normally.

[2]

Markscheme

ALTERNATIVE 1

$$\text{calculates total current} = \frac{35}{15} \times 8 \ll = 18.7 \text{ A} \gg \checkmark$$

$$\text{resistance} = \ll \frac{15}{18.7} = \gg 0.80 \text{ «}\Omega\text{» } \checkmark$$

ALTERNATIVE 2

$$\text{calculates total power} = 35 \times 8 \ll = 280 \text{ W} \gg \checkmark$$

$$\text{resistance} = \ll \frac{15^2}{280} = \gg 0.80 \text{ «}\Omega\text{» } \checkmark$$

ALTERNATIVE 3

$$\text{calculates individual resistance} = \frac{15^2}{35} \ll = 6.43 \Omega \gg \checkmark$$

$$\text{resistance} = \ll \frac{6.43}{8} \gg = 0.80 \text{ «}\Omega\text{» } \checkmark$$

(a.iii) Calculate the current in the primary of the transformer assuming that it is ideal.

[2]

Markscheme

$$\text{total power required} = 280 \text{ «W»}$$

OR

$$\text{uses factor } \frac{3300}{450}$$

OR

$$\text{total current} = 18.7 \text{ «A» } \checkmark$$

$$\text{current} = 2.5 \text{ OR } 2.6 \text{ «A» } \checkmark$$

Award **[2] marks** for a bald correct answer.

Allow **ECF** from (a)(iii).

- (a.iv) Flux leakage is one reason why a transformer may not be ideal. Explain the effect of flux leakage on the transformer.

[2]

Markscheme

the secondary coil does not enclose all flux «lines from core» ✓

induced emf in secondary

OR

power transferred to the secondary

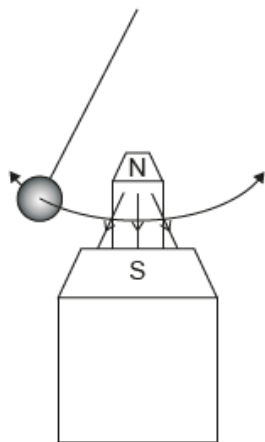
OR

efficiency is less than expected ✓

Award **[0]** for references to eddy currents/heating of the core as the reason.

Award **MP2** if no reason stated.

- (b) A pendulum with a metal bob comes to rest after 200 swings. The same pendulum, released from the same position, now swings at 90° to the direction of a strong magnetic field and comes to rest after 20 swings.



Explain why the pendulum comes to rest after a smaller number of swings.

[4]

Markscheme

bob cuts mag field lines

OR

there is a change in flux linkage ✓

induced emf across bob ✓

leading to eddy/induced current in bob ✓

eddy/induced current produces a magnetic field that opposes «direction of» motion ✓

force due to the induced magnetic field decelerates bob ✓

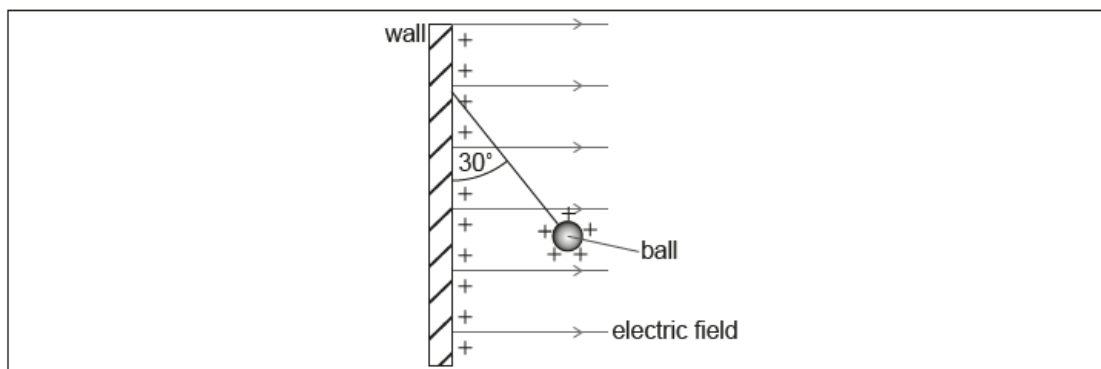
damping of pendulum increases/there is additional «magnetic» damping ✓

MP4 and MP5 can be expressed in terms of energy transfer from kinetic energy of bob to electrical/thermal energy in bob

11. [Maximum mark: 15]

21M.2.HL.TZ2.3

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



- (a) The charge per unit area on the surface of the wall is σ . It can be shown that the electric field strength E due to the charge on the wall is given by the equation

$$E = \frac{\sigma}{2\epsilon_0}.$$

Demonstrate that the units of the quantities in this equation are consistent.

[2]

Markscheme

identifies units of σ as C m^{-2} ✓

$$\frac{C}{m^2} \times \frac{Nm^2}{C^2} \text{ seen and reduced to } \text{N C}^{-1} \checkmark$$

Accept any analysis (eg dimensional) that yields answer correctly

- (b.i) The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg.

Determine the horizontal force that acts on the ball.

[3]

Markscheme

$$\text{horizontal force } F \text{ on ball} = T \sin 30 \checkmark$$

$$T = \frac{mg}{\cos 30} \checkmark$$

$$F \llcorner = mg \tan 30 = 0.025 \times 9.8 \times \tan 30 \llcorner = 0.14 \llcorner \text{N} \llcorner \checkmark$$

$$\text{Allow } g = 10 \text{ N kg}^{-1}$$

Award **[3] marks** for a bald correct answer.

Award **[1max]** for an answer of zero, interpreting that the horizontal force refers to the horizontal component of the net force.

- (b.ii) The charge on the ball is $1.2 \times 10^{-6} \text{ C}$. Determine σ .

[2]

Markscheme

$$E = \frac{0.14}{1.2 \times 10^{-6}} \llcorner = 1.2 \times 10^5 \llcorner \checkmark$$

$$\sigma = \llcorner \frac{2 \times 8.85 \times 10^{-12} \times 0.14}{1.2 \times 10^{-6}} \ggcorner = 2.1 \times 10^{-6} \llcorner \text{C m}^{-2} \ggcorner \checkmark$$

Allow **ECF** from the calculated F in (b)(i)

Award **[2]** for a bald correct answer.

- (c) The thread breaks. Explain the initial subsequent motion of the ball.

[3]

Markscheme

horizontal/repulsive force and vertical force/pull of gravity act on the ball ✓

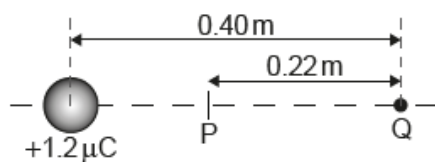
so ball has constant acceleration/constant net force ✓

motion is in a straight line ✓

at 30° to vertical away from wall/along original line of thread ✓

The centre of the ball, still carrying a charge of $1.2 \times 10^{-6} \text{ C}$, is now placed 0.40 m from a point charge Q. The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero. The distance PQ is 0.22 m.



- (d.i) Calculate the charge on Q. State your answer to an appropriate number of significant figures.

[3]

Markscheme

$$\frac{Q}{0.22^2} = \frac{1.2 \times 10^{-6}}{0.18^2} \quad \checkmark$$

$$\ll + \gg 1.8 \times 10^{-6} \ll \text{C} \gg \quad \checkmark$$

2sf ✓

*Do not award **MP2** if charge is negative*

*Any answer given to 2 sig figs scores **MP3***

- (d.ii) Outline, without calculation, whether or not the electric potential at P is zero.

[2]

Markscheme

work must be done to move a «positive» charge from infinity to P «as both charges are positive»

OR

reference to both potentials positive and added

OR

identifies field as gradient of potential and with zero value ✓

therefore, point P is at a positive / non-zero potential ✓

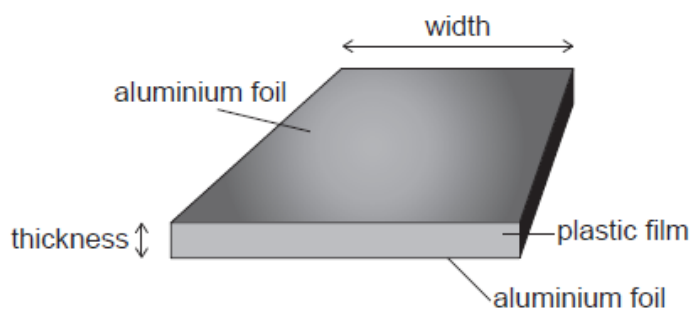
Award [0] for bald answer that P has non-zero potential

12. [Maximum mark: 12]

19M.2.HL.TZ1.8

A student makes a parallel-plate capacitor of capacitance 68 nF from aluminium foil and plastic film by inserting one sheet of plastic film between two sheets of aluminium foil.

diagram not to scale



The aluminium foil and the plastic film are 450 mm wide.

The plastic film has a thickness of 55 μm and a permittivity of $2.5 \times 10^{-11} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

(a.i) Calculate the total length of aluminium foil that the student will require.

[3]

Markscheme

$$\text{length} = \frac{d \times C}{\text{width} \times \epsilon} \quad \checkmark$$

$$= 0.33 \text{ «m»} \quad \checkmark$$

$$\text{so } 0.66/0.67 \text{ «m» «as two lengths required»} \quad \checkmark$$

Examiners report

Many candidates were able to use the proper equation to calculate the length of one piece of aluminum foil for the first two marks, but very few doubled the length for the final mark.

- (a.ii) The plastic film begins to conduct when the electric field strength in it exceeds 1.5 MN C^{-1} . Calculate the maximum charge that can be stored on the capacitor.

[2]

Markscheme

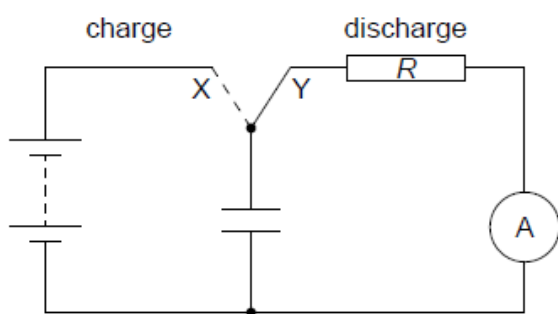
$$1.5 \times 10^6 \times 55 \times 10^{-6} = 83 \text{ «V»} \checkmark$$

$$q \ll CV = 5.6 \times 10^{-6} \text{ «C»} \checkmark$$

Examiners report

This question was challenging for many candidates. While some candidates were able to use proper equations for capacitors to determine the charge some of the candidates attempted to use electrostatic equations for the electric field around a point charge to solve this problem.

The student uses a switch to charge and discharge the capacitor using the circuit shown. The ammeter is ideal.



The emf of the battery is 12 V.

- (b.i) The resistor R in the circuit has a resistance of $1.2 \text{ k}\Omega$. Calculate the time taken for the charge on the capacitor to fall to 50 % of its fully charged value.

[3]

Markscheme

$$0.5 = e^{-\frac{t}{RC}} = e^{-\frac{t}{1200 \times 6.8 \times 10^{-8}}}$$

$$t = \leftarrow 1200 \times 6.8 \times 10^{-8} \times \ln 0.5 \checkmark$$

$$5.7 \times 10^{-5} \text{ «s» } \checkmark$$

OR

$$\text{use of } t \frac{1}{2} = RC \times \ln 2 \checkmark$$

$$1200 \times 6.8 \times 10^{-8} \times 0.693 \checkmark$$

$$5.7 \times 10^{-5} \text{ «s» } \checkmark$$

Examiners report

This question was also challenging for many candidates, with not an insignificant number leaving it blank. The candidates who did attempt it generally set up a correct equation, but ran into some simple calculation and power of ten errors. Some candidates attempted to solve the equation using basic circuit equations, which did not receive any marks.

- (b.ii) The ammeter is replaced by a coil. Explain why there will be an induced emf in the coil while the capacitor is discharging.

[2]

Markscheme

mention of Faraday's law $\checkmark$

indicating that changing current in discharge circuit leads to change in flux in coil/change in magnetic field «and induced emf» $\checkmark$

Examiners report

This is an explain question, so there was an expectation for a fairly detailed response. Many candidates missed the fact that the discharging capacitor is causing the current in the coil to change in time, and that this is what is inducing the emf in the coil. Many simply stated that the current created a magnetic field with not complete explanation of induction.

- (b.iii) Suggest **one** change to the discharge circuit, apart from changes to the coil, that will increase the maximum induced emf in the coil.

[2]

Markscheme

decrease/reduce ✓

resistance (R) **OR** capacitance (C) ✓

Examiners report

Candidates who recognized that something about the discharge circuit (not the charging circuit) needed to be changed generally suggested that something had to change with the resistance or capacitance. It should be noted that even though this was the last question on the exam, it was attempted at a higher rate than many of the other questions on the exam.

13. [Maximum mark: 10]

19M.2.HL.TZ1.5

The moon Phobos moves around the planet Mars in a circular orbit.

(a.i) Outline the origin of the force that acts on Phobos.

[1]

Markscheme

gravitational attraction/force/field «of the planet/Mars» ✓

Do not accept "gravity".

Examiners report

This was generally well answered, although some candidates simply used the vague term "gravity" rather than specifying that it is a gravitational force or a gravitational field. Candidates need to be reminded about using proper physics terms and not more general, "every day" terms on the exam.

(a.ii) Outline why this force does no work on Phobos.

[1]

Markscheme

the force/field and the velocity/displacement are at 90° to each other **OR**

there is no change in GPE of the moon/Phobos ✓

Examiners report

Some candidates connected the idea that the gravitational force is perpendicular to the velocity (and hence the displacement) for the mark. It was also allowed to discuss that there is no change in gravitational potential energy, so therefore no work was being done. It was not acceptable to simply state that the net displacement over one full orbit is zero. Unfortunately, some candidates suggested that there is no net force on the moon so there is no work done, or that the moon is so much smaller so no work could be done on it.

- (b.i) The orbital period T of a moon orbiting a planet of mass M is given by

$$\frac{R^3}{T^2} = kM$$

where R is the average distance between the centre of the planet and the centre of the moon.

Show that $k = \frac{G}{4\pi^2}$

[3]

Markscheme

ALTERNATE 1

«using fundamental equations»

use of Universal gravitational force/acceleration/orbital velocity equations ✓

equating to centripetal force or acceleration. ✓

rearranges to get $k = \frac{G}{4\pi^2}$ ✓

ALTERNATE 2

«starting with $\frac{R^3}{T^2} = kM$ »

substitution of proper equation for T from orbital motion equations ✓

substitution of proper equation for M **OR** R from orbital motion equations ✓

rearranges to get $k = \frac{G}{4\pi^2}$ ✓

Examiners report

This was another “show that” derivation. Many candidates attempted to work with universal gravitation equations, either from memory or the data booklet, to perform this derivation. The variety of correct solution paths was quite impressive, and many candidates who attempted this question were able to receive some marks. Candidates should be reminded on “show that” questions that it is never allowed to work backwards from the given answer. Some candidates also made up equations (such as $T = 2\pi r$) to force the derivation to work out.

- (b.ii) The following data for the Mars–Phobos system and the Earth–Moon system are available:

Mass of Earth = 5.97×10^{24} kg

The Earth–Moon distance is 41 times the Mars–Phobos distance.

The orbital period of the Moon is 86 times the orbital period of Phobos.

Calculate, in kg, the mass of Mars.

[2]

Markscheme

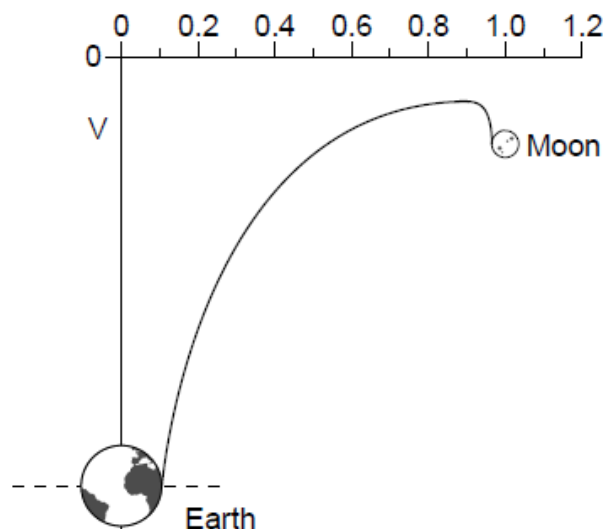
$$m_{\text{Mars}} = \left(\frac{R_{\text{Mars}}}{R_{\text{Earth}}} \right)^3 \left(\frac{T_{\text{Earth}}}{T_{\text{Mars}}} \right)^2 m_{\text{Earth}} \text{ or other consistent re-arrangement } \checkmark$$

6.4×10^{23} «kg» $\checkmark$

Examiners report

This question was challenging for candidates. The candidates who started down the correct path of using the given derived value from 5bi often simply forgot that the multiplication factors had to be squared and cubed as well as the variables.

- (c) The graph shows the variation of the gravitational potential between the Earth and Moon with distance from the centre of the Earth. The distance from the Earth is expressed as a fraction of the total distance between the centre of the Earth and the centre of the Moon.



[3]

Determine, using the graph, the mass of the Moon.

Markscheme

read off separation at maximum potential 0.9 ✓

equating of gravitational field strength of earth and moon at that location *OR* $\frac{M_E}{0.9^2} = \frac{M_M}{0.1^2}$ ✓

7.4×10^{22} «kg» ✓

Allow ECF from MP1

Examiners report

This question was left blank by many candidates, and very few who attempted it were able to successfully recognize that the gravitational fields of the Earth and Moon balance at 0.9r and then use the proper equation to calculate the mass of the Moon.

14. [Maximum mark: 4]

19M.2.HL.TZ2.10

A small magnet is dropped from rest above a stationary horizontal conducting ring. The south (S) pole of the magnet is upwards.

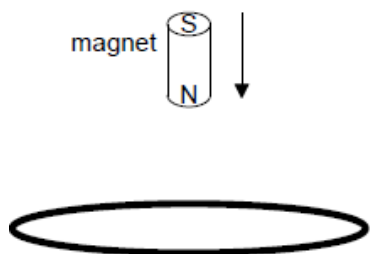


Diagram 1: side view

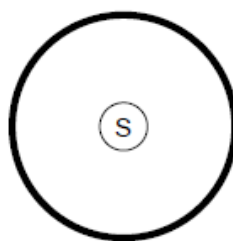


Diagram 2: view from above

- (a) While the magnet is moving towards the ring, state why the magnetic flux in the ring is increasing.

[1]



Markscheme

the magnetic field at the position of the ring is increasing «because the magnet gets closer to the ring» ✓

Examiners report

This was well-answered.

- (b) While the magnet is moving towards the ring, sketch, using an arrow on **Diagram 2**, the direction of the induced current in the ring.

[1]



Markscheme

the current must be counterclockwise «in diagram 2» ✓

eg:

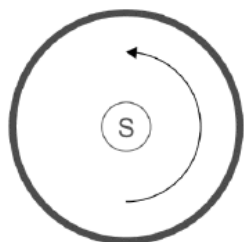


Diagram 2: view from above

Examiners report

Answers here were reasonably evenly split between clockwise and anti-clockwise, with the odd few arrows pointing left or right.

- (c) While the magnet is moving towards the ring, deduce the direction of the magnetic force on the magnet.

[2]



Markscheme

since the induced magnetic field is upwards

OR

by Lenz law the change «of magnetic field/flux» must be opposed

OR

by conservation of energy the movement of the magnet must be opposed ✓

therefore the force is repulsive/upwards ✓

Examiners report

The majority of candidates recognised that the magnetic force would be upwards and the most common way of explaining this was via Lenz's law. Students needed to get across that the force is opposing a change or a motion.

15. [Maximum mark: 6]

19M.2.HL.TZ2.9

A planet of mass m is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is V .

- (ai) Show that the total energy of the planet is given by the equation shown.

$$E = \frac{1}{2}mV$$

[2]



Markscheme

$$E = \frac{1}{2}m \frac{GM}{r} - \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r} \quad \checkmark$$

$$\text{comparison with } V = -\frac{GM}{r} \quad \checkmark$$

«to give answer»

Examiners report

This was generally well answered but with candidates sometimes getting in to trouble over negative signs but otherwise producing well-presented answers.

- (aii) Suppose the star could contract to half its original radius without any loss of mass.
Discuss the effect, if any, this has on the total energy of the planet.

[2]



Markscheme

ALTERNATIVE 1

«at the position of the planet» the potential depends only on the mass of the star /does not depend on the radius of the star ✓

the potential will not change and so the energy will not change ✓

ALTERNATIVE 2

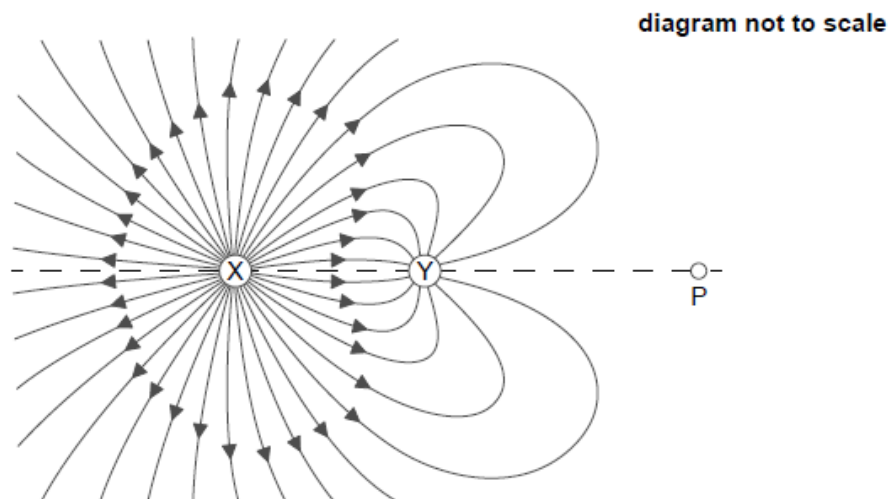
r / distance between the centres of the objects / orbital radius remains unchanged ✓

since $E_{Total} = -\frac{1}{2} \frac{GMm}{r}$, energy will not change ✓

Examiners report

A large number of candidates thought that the total energy of the planet would change, mostly double.

- (b) The diagram shows some of the electric field lines for two fixed, charged particles X and Y.



[2] □

The magnitude of the charge on X is Q and that on Y is q . The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

At P the electric field is zero. Determine, to **one** significant figure, the ratio $\frac{Q}{q}$.

Markscheme

$$\frac{kQ}{(0.600+0.820)^2} = \frac{kq}{0.820^2} \quad \checkmark$$

$$\frac{Q}{q} = \ll \frac{(0.600+0.820)^2}{0.820^2} = 2.9988 \approx \gg 3 \quad \checkmark$$

Examiners report

The majority of candidates had an idea of the basic technique here but it was surprisingly common to see the squared missing from the expression for field strengths.

16. [Maximum mark: 4]

19M.2.HL.TZ2.10

A small magnet is dropped from rest above a stationary horizontal conducting ring. The south (S) pole of the magnet is upwards.

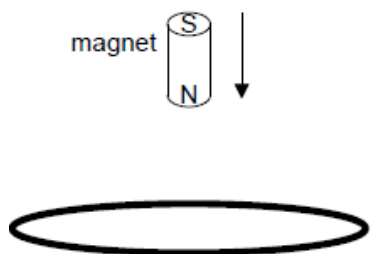


Diagram 1: side view

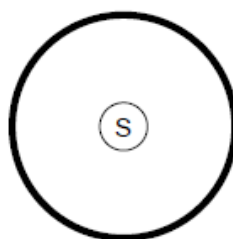


Diagram 2: view from above

- (a) While the magnet is moving towards the ring, state why the magnetic flux in the ring is increasing.

[1]



Markscheme

the magnetic field at the position of the ring is increasing «because the magnet gets closer to the ring» ✓

Examiners report

This was well-answered.

- (b) While the magnet is moving towards the ring, sketch, using an arrow on **Diagram 2**, the direction of the induced current in the ring.

[1]



Markscheme

the current must be counterclockwise «in diagram 2» ✓

eg:

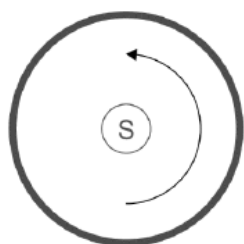


Diagram 2: view from above

Examiners report

Answers here were reasonably evenly split between clockwise and anti-clockwise, with the odd few arrows pointing left or right.

- (c) While the magnet is moving towards the ring, deduce the direction of the magnetic force on the magnet.

[2]



Markscheme

since the induced magnetic field is upwards

OR

by Lenz law the change «of magnetic field/flux» must be opposed

OR

by conservation of energy the movement of the magnet must be opposed ✓

therefore the force is repulsive/upwards ✓

Examiners report

The majority of candidates recognised that the magnetic force would be upwards and the most common way of explaining this was via Lenz's law. Students needed to get across that the force is opposing a change or a motion.

17. [Maximum mark: 6]

19M.2.HL.TZ2.9

A planet of mass m is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is V .

- (ai) Show that the total energy of the planet is given by the equation shown.

[2]



$$E = \frac{1}{2}mV$$

Markscheme

$$E = \frac{1}{2}m \frac{GM}{r} - \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r} \quad \checkmark$$

comparison with $V = -\frac{GM}{r} \quad \checkmark$

«to give answer»

Examiners report

This was generally well answered but with candidates sometimes getting in to trouble over negative signs but otherwise producing well-presented answers.

- (aii) Suppose the star could contract to half its original radius without any loss of mass.
Discuss the effect, if any, this has on the total energy of the planet.

[2]



Markscheme

ALTERNATIVE 1

«at the position of the planet» the potential depends only on the mass of the star /does not depend on the radius of the star ✓

the potential will not change and so the energy will not change ✓

ALTERNATIVE 2

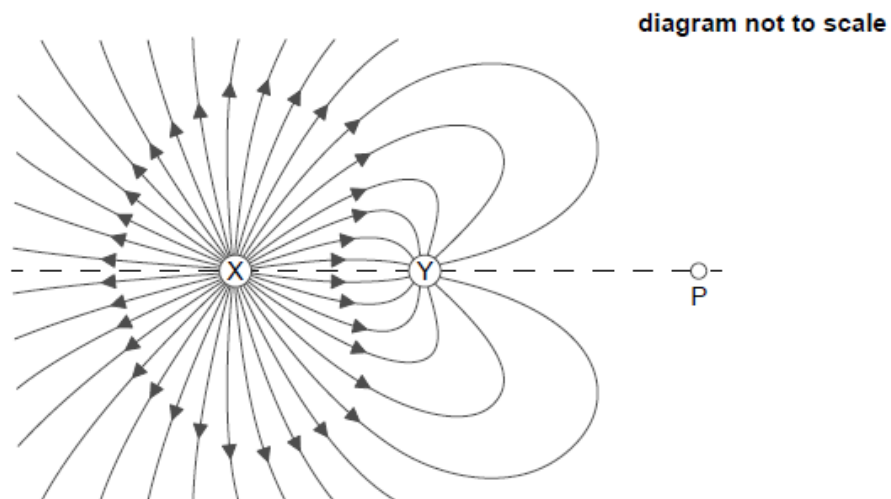
r / distance between the centres of the objects / orbital radius remains unchanged ✓

since $E_{Total} = -\frac{1}{2} \frac{GMm}{r}$, energy will not change ✓

Examiners report

A large number of candidates thought that the total energy of the planet would change, mostly double.

- (b) The diagram shows some of the electric field lines for two fixed, charged particles X and Y.



[2] □

The magnitude of the charge on X is Q and that on Y is q . The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

At P the electric field is zero. Determine, to **one** significant figure, the ratio $\frac{Q}{q}$.

Markscheme

$$\frac{kQ}{(0.600+0.820)^2} = \frac{kq}{0.820^2} \quad \checkmark$$

$$\frac{Q}{q} = \ll \frac{(0.600+0.820)^2}{0.820^2} = 2.9988 \approx \gg 3 \quad \checkmark$$

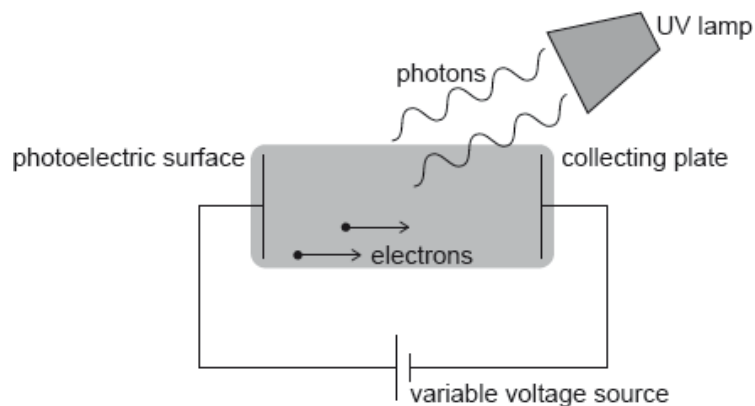
Examiners report

The majority of candidates had an idea of the basic technique here but it was surprisingly common to see the squared missing from the expression for field strengths.

18. [Maximum mark: 11]

18M.2.HL.TZ1.8

Hydrogen atoms in an ultraviolet (UV) lamp make transitions from the first excited state to the ground state. Photons are emitted and are incident on a photoelectric surface as shown.



(a) Show that the energy of photons from the UV lamp is about 10 eV.

[2]

Markscheme

$$E_1 = -13.6 \text{ «eV»} \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ «eV»}$$

energy of photon is difference $E_2 - E_1 = 10.2 \approx 10 \text{ eV}$

Must see at least 10.2 eV.

[2 marks]

The photons cause the emission of electrons from the photoelectric surface. The work function of the photoelectric surface is 5.1 eV.

(b.i) Calculate, in J, the maximum kinetic energy of the emitted electrons.

[2]

Markscheme

$$10 - 5.1 = 4.9 \text{ «eV»}$$

$$4.9 \times 1.6 \times 10^{-19} = 7.8 \times 10^{-19} \text{ «J»}$$

Allow 5.1 if 10.2 is used to give $8.2 \times 10^{-19} \text{ «J»}$.

(b.ii) Suggest, with reference to conservation of energy, how the variable voltage source can be used to stop all emitted electrons from reaching the collecting plate.

[2]

Markscheme

EPE produced by battery

exceeds maximum KE of electrons / electrons don't have enough KE

For first mark, accept explanation in terms of electric potential energy difference of electrons between surface and plate.

[2 marks]

- (b.iii) The variable voltage can be adjusted so that no electrons reach the collecting plate.
Write down the minimum value of the voltage for which no electrons reach the collecting plate.

[1]

Markscheme

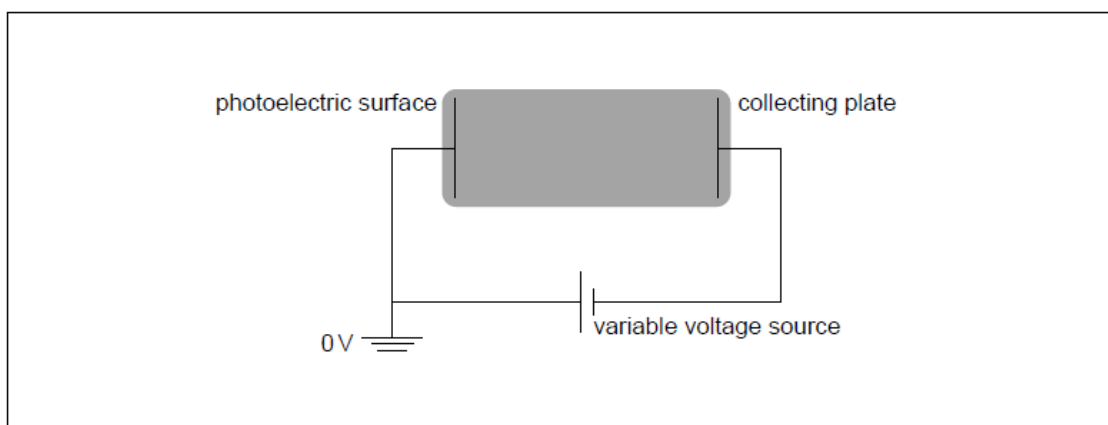
4.9 «V»

Allow 5.1 if 10.2 is used in (b)(i).

Ignore sign on answer.

[1 mark]

The electric potential of the photoelectric surface is 0V. The variable voltage is adjusted so that the collecting plate is at -1.2 V .



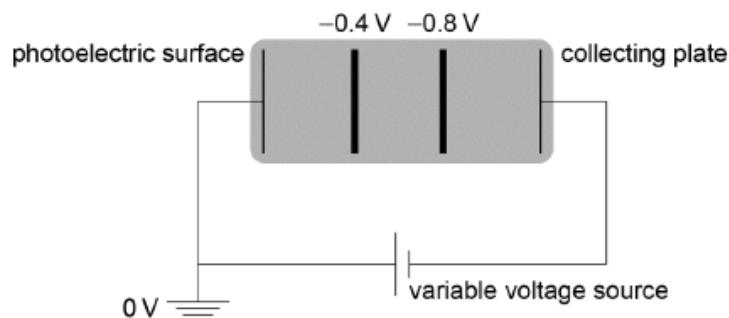
- (c.i) On the diagram, draw and label the equipotential lines at -0.4 V and -0.8 V .

[2]

Markscheme

two equally spaced vertical lines (judge by eye) at approximately 1/3 and 2/3

labelled correctly



[2 marks]

- (c.ii) An electron is emitted from the photoelectric surface with kinetic energy 2.1 eV.
Calculate the speed of the electron at the collecting plate.

[2]

Markscheme

kinetic energy at collecting plate = 0.9 «eV»

$$\text{speed} = \sqrt{\frac{2 \times 0.9 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}}} = 5.6 \times 10^5 \text{ «ms}^{-1}\text{»}$$

Allow ECF from MP1

[2 marks]

19. [Maximum mark: 11]

18M.2.HL.TZ2.6

A planet has radius R . At a distance h above the surface of the planet the gravitational field strength is g and the gravitational potential is V .

- (a.i) State what is meant by gravitational field strength.

[1]

Markscheme

the «gravitational» force per unit mass exerted on a point/small/test mass

[1 mark]

(a.ii) Show that $V = -g(R + h)$.

[2]

Markscheme

at height h potential is $V = -\frac{GM}{(R+h)}$

field is $g = \frac{GM}{(R+h)^2}$

«dividing gives answer»

Do not allow an answer that starts with $g = -\frac{\Delta V}{\Delta r}$ and then cancels the deltas and substitutes $R + h$

[2 marks]

(a.iii) Draw a graph, on the axes, to show the variation of the gravitational potential V of the planet with height h above the surface of the planet.

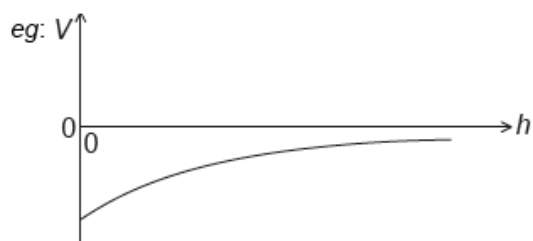


[2]

Markscheme

correct shape and sign

non-zero negative vertical intercept



[2 marks]

- (b) A planet has a radius of 3.1×10^6 m. At a point P a distance 2.4×10^7 m above the surface of the planet the gravitational field strength is 2.2 N kg^{-1} . Calculate the gravitational potential at point P, include an appropriate unit for your answer.

[1]

Markscheme

$$V = \left\langle -2.2 \times (3.1 \times 10^6 + 2.4 \times 10^7) \right\rangle \Rightarrow \left\langle - \right\rangle 6.0 \times 10^7 \text{ J kg}^{-1}$$

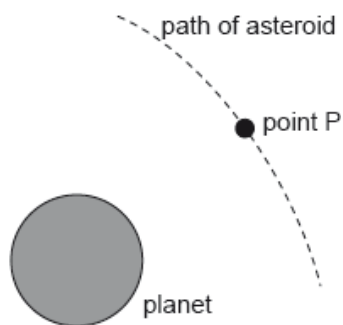
Unit is essential

Allow eg MJ kg⁻¹ if power of 10 is correct

Allow other correct SI units eg m²s⁻², N m kg⁻¹

[1 mark]

- (c) The diagram shows the path of an asteroid as it moves past the planet.



When the asteroid was far away from the planet it had negligible speed. Estimate the speed of the asteroid at point P as defined in (b).

[3]

Markscheme

total energy at P = 0 / KE gained = GPE lost

$$\llcorner \frac{1}{2}mv^2 + mV = 0 \Rightarrow \llcorner v = \sqrt{-2V}$$

$$v = \llcorner \sqrt{2 \times 6.0 \times 10^7} = \llcorner 1.1 \times 10^4 \text{ «ms}^{-1}\text{»}$$

Award [3] for a bald correct answer

Ignore negative sign errors in the workings

Allow ECF from 6(b)

[3 marks]

- (d) The mass of the asteroid is 6.2×10^{12} kg. Calculate the gravitational force experienced by the **planet** when the asteroid is at point P.

[2]

Markscheme

ALTERNATIVE 1

force on asteroid is $\llcorner 6.2 \times 10^{12} \times 2.2 = \llcorner 1.4 \times 10^{13} \text{ «N»}$

$\llcorner$ by Newton's third law» this is also the force on the planet

ALTERNATIVE 2

mass of planet = 2.4×10^{25} «kg» $\llcorner$ from $V = -\frac{GM}{(R+h)}$ »

force on planet $\llcorner \frac{GMm}{(R+h)^2} = 1.4 \times 10^{13} \text{ «N»}$

MP2 must be explicit

[2 marks]